Supplementary file number 3 (S3): Quality Assessment of Studies.

cochrane risk of bias		Ottenheim 2018		Lynn 2018		Dalley 2016		Pontes 2015		Leonhardt 2014		Burns 2013		Brown 2012		Burns Nader 2011		Victorine 2002		Yuval 2008	
entry		Judgement	Support for judgment	Judgement	Support for judgment	Judgement	Support for judgment	Judgement	Support for judgment	Judgement	Support for judgment	Judgement	Support for judgment	Judgement	Support for judgment	Judgement	Support for judgment	Judgement	Support for judgment	Judgement	Support for judgment
Random sequence generation (selection bias)	R1	High Risk	No randomisation - 794 kindergarten children (age 4-7) participated in the Teddy Bear Hospital at Leiden University Medical Centre (Leiden, the Netherlands). Data was collected on 543 children attending elementary schools in the Leiden region.	High Risk	No randomisation - cross sectional descriptive study of children taking part in TBH sessions between 28 March and 20 May 2016. All children aged 5 to 8 years old were recruited from kindergartens and primary schools which	High Risk	No randomisation - A convenience sample was recruited from groups of children already scheduled to attend a Teddy Bear Clinic Tour at the Hill's Pet Nutrition Primary Healthcare Centre of the Ontario Veterinary College (OVC)	High Risk	No randomisation - subjects were not randomly selected, and therefore this is not a true experimental study.	High Risk	No randomisation - Randomization was not possible because the TBH is an established local project and any kindergarten in the area is free to decide whether it will take part or not.	Unclear Risk	The study did not specify how randomisation was done - Child participants were randomly assigned to 1 of 4 groups.	Unclear Risk	The study did not specify how randomisation was done - Children were randomly assigned to participate in a medical collage activity or a control group.	Low Risk	Assignment to groups was determined using random sample numbers generated by a computer software program	High Risk	No randomisation - We estimated a total of about 124 children attended this Teddy Bear Hospital. Every child was given a questionnaire to fill-in prior to attending Teddy Bear Hospital and after the experience.	High Risk	No randomisation - Forty-one preschool children, age 3–6.5 years from three kindergartens in Beer Sheva, Israel, took part in this study. Fifty preschool children, age matched and from a similar residential area, served as the control group.
	R2	High Risk	Participants were selected based on their attendance to teddy bear hospital. No randomisation	High Risk	All the children from all the schools and kindergardens that has participated in TBH were recruited with the exception of children without parental consent. No randomisation.	High Risk	A convenience sample was recruited from groups of children already scheduled to attend a Teddy Bear Clinic Tour at the Hill's Pet Nutrition Primary Healthcare CentreoftheOntarioVeterinaryCollege (OVC).	High Risk	subjects were not randomly selected, and therefore this is not a true experimental study. No randomization	High Risk	Randomization was not possible because the TBH is an established local project and any kindergarten in the area is free to decide whether it will take part or not. No randomization	High Risk	Seventy-two caregiver–child dyads were recruited to participate, following institutional review board approval. The children (n = 33 males) were attending a scheduled, or walk-in, doctor's appointment at a pediatric clinic in a city in the southeast region of the United States, for either a well checkup (ie, school physical, vaccinations, etc) or for an ill checkup (ie, respiratory infection, skin rash, etc). No randomisation.	Unclear Risk	A total of 30 children ranging in age from three to five were recruited with parental consent and participant (verbal) assent from a pediatric clinic in Tuscaloosa, AL. But there was no explanantion on the mode of selection.	High Risk	No randomisation. The children (n= 33 males) were attending a scheduled, or walk-in, doctor's appointment at a general medical school pediatric clinic.	High Risk	No randomisation. As a child comes in, a teddy doctor in the reception area will register the child and help them fill in part one of the questionnaire. Participants are selected based on attendance.	High Risk	No randomisation. Forty-one preschool children, age 3–6.5 years (mean 5.1 ±0.7 years), from three kindergartens in Beer Sheva, Israel, took part in this study. Fifty preschool children, age matched and from a similar residential area, served as the control group.
Allocation concealment (selection bias)	R1	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	Unclear Risk	Method of concealment was not described - Children were distributed to an experimental group, with 30 children taking part in an instructional therapeutic play session before vaccination, and to a control group, with 30 children who did not receive any kind of previous emotional preparation.	High Risk	No allocation concealment - experimental group was recruited from kindergarten who registered to participate in TBH and control group from non-participating facilities	High Risk	Method of concealment was not described	High Risk	Method of concealment was not described	High Risk	Method of concealment was not described	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - intervention group was recruited from kindergarten who participated in TBH and control group from a similar residential area
	R2	High Risk	The same questions were used thrice, from pre exposure to TBH, to post TBH lecture and post TBH participation. No allocation concealment.	Low risk	Each children were assigned a coded number without identifier.	Unclear risk	The allocation concealment method was not sufficiently described.	High Risk	Data were recorded in a form with information about identification of child and the reactions during vaccination. No allocation concealment was practiced	Low Risk	Children were assessed in small groups (up to 5 children) in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. No individual identifiers involved.	Low Risk	Child participants were randomly assigned to 1 of 4 groups: (a) a medical play group, where children were given a doll and authentic medical materials to play with (ie, stethoscope, otoscope, tongue depressor, etc); (b) a medical information video group, in which children were shown a taped medical play session; (c) a typical play group, where children played a board game; or (d) a control group whose participants viewed a video on safari life.	Low Risk	The children were randomly assigned to participate in medical collage activity or control group.	Low Risk	Child participants were randomly assigned to one of four groups: 1) a medical play group, where children were given a doll and authentic medical equipment, 2) a medical information video group, in which children were shown a taped medical play session, 3) a typical play group (i.e., played Connect Four), or 4) a control group whose participants viewed a video on safari life. Assignment to groups was determined using random sample numbers generated by a computer software program	Low Risk	All the questionnaires are numbered which means each pariticipants are allocated a number and identifiers are not used.	Low Risk	The paticipants are divivded into intervention group and control group.
Blinding of participants and personnel (performance bias)	R1	High Risk	No blinding - arranged for all participating schools a box with materials and a concrete lesson for the teachers with the intention that this lesson was given before the visit to the Teddy Bear Hospital.	High Risk	No blinding - Questionnaire was administered right before TBH and immediately after TBH	High Risk	No blinding - Data were collected from participating children immediately before and immediately afer the tour	High Risk	Blinding is not posible - children in experimental group have to participate in instructional therapeutic play sessions.	High Risk	No blinding - Children were assessed in small groups in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. Control children were interviewed twice as well and at a similar time interval.	High Risk	Blinding is not possible - children were either involved in a medical play group, a medical information video group, a typical play group or a control group whose participants viewed a video on safari life.	High Risk	Blinding is not possible - study group has to participate in medical collage activity	High Risk	Blinding is not possible - children were either involveld in a medical play group, a medical information video group, a typical play group or a control group whose participants viewed a video on safari life.	High Risk	No blinding - Every child was given a questionnaire to fill-in prior to attending Teddy Bear Hospital and after the experience.	High Risk	No blinding - a simple one-item visual analog scale of anxiety about hospitalization was assessed individually one day prior to the intervention and again a week after the intervention when the children were in kindergarten.
	R2	High Risk	No Blindng was applied. All the participants of TBH was included in selection with a prior lesson before participation. Despite that, a significant improvement was noticed only after the intervention which is T2. For question one, significant difference were seen between T2 and T1, T2 and T0, but no difference were seen between T1 and T0. Outcome is not influenced by the lack of blinding.	High Risk	A lesson regarding the concept of TBH was provided to children immediately after TBH session to assess knowledge acquired and their change of feelings towards healthcare.	High Risk	The data was collected from participants before and immediately after the tour.	High Risk	After informing children and their legal guardians about the study and their participation, those selected for the experimental group were invited to play. No blinding	High Risk	Children were assessed in small groups (up to 5 children) in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. No individual identifiers involved. Control children were interviewed twice as well and at a similar time interval (about 21 days). No blinding	Low Risk	Upon participating in their assigned activity, the children underwent a doctor's checkup where a trained nurse will conduct a behavioural assessment on the children.	Low Risk	A behavioural Observation Scale was used by researchers to evaluate the child's behaviour pre and post study group and control group acitvity participation.	Low Risk	The child's pulse reading (Indication of anxiety) and Behaviour rating were conducted before and after the group activities were conuducted in the clinical exam room.	High Risk	The child participants were required to fill in 2 questionairres pre and post intervention. No blinding was applied.	Low Risk	Assessment included a simple one-item visual analog scale of anxiety about hospitalization. The scale included five facial expressions indicating a happy face on one side of the scale and a very distressed face on the other. Children had to choose the face that best described the way they would feel if they were about to be hospitalized [7]. This was assessed individually one day prior to the intervention and again a week after the intervention when the children were in kindergarten.
Blinding of outcome assessment(detection bias) (patient-reported outcomes)	R1	Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk	No mention about blinding - Two undergraduate research assistants independently coded all of the participants' responses. Disagreements were resolved by discussion and consensus.	High Risk	Reactions were observed by researcher.	Unclear Risk	No mention about blinding - The interviews with the children were conducted by two trained students who were supervised by an experienced psychologist.	Low Risk	The nurses who rated the children's behaviour were unaware of the group assignments	High Risk	Reactions were observed by researcher. (Behaviour Observation Scale)	Low Risk	A researcher, blind to the children's group assignment, rated the children's affect and behaviors. The developmental interview was administered to the parent by a researcher blind to the children's group assignment.	Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk	The study did not address blinding of outcome assessors
	R2	Unclear risk	Insufficient details for judgement	Unclear Risk	The study did not adress this outcome.	Unclear risk	Insufficient details for judgement	High Risk	The children's reaction were observed by the researches themselve while vaccinating.	High Risk	The interviews with the children were conducted by two trained students who were supervised by an experienced psychologist.	High Risk	Because there will be trained nurse assigned to conduct the behavioural assessment on the children.	High Risk	The researchers evaluate the children's behaviour by using Behavioural Observation Scale.	Low Risk	A researcher, blind to the children's group assignment, rated the children's affect and behaviors before the assigned activity, during triage, and after the doctor's visit.	Low Risk	Questionairres were used to assess the effects of TBH to reduce child's fear of doctors and hospitals.	Unclear Risk	Insifficient details for judgement
Blinding of outcome assessment (detection bias) (all-cause mortality)	R1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	R2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Incomplete outcome data addressed (attrition bias) (short-term 2-6 weeks)	R1	High Risk	The four questions of the questionnaire were analysed separately so no effect on the interpretation of the results are expected however missing data percentage ranges from 17.5%-19.89%. Data analysis was done only for completed questionnaire.	High Risk	We excluded children who had incomplete data, such as missing answers to any of the multiple choice question in the survey. Children who completed the multiple choice component but did not answer the open-ended questions were still included. (Excluded 82 out of 334 - 24.55%)	Low Risk	A total of 71 participants had parental consent and assented to participate; no data were excluded from the final analysis.	Low Risk	No data were excluded from the final analysis	Low Risk	For reasons of non-appropriate age and outlier data, 8 children were excluded from final analysis. (<15%)	Low Risk	No data were excluded from the final analysis	Low Risk	No data were excluded from the final analysis	Low Risk	No data were excluded from the final analysis	High Risk	Total 124 questionnaires, questionnaires numbered 86-124 were not found. Out of the 85 questionnaires, 28 of the questionnaires were not filled in fully. Therefore, only 57 fully completed questionnaires were considered for this study. (Excluded 67 out of 124 - >15%)	Low Risk	Three children in the intervention group (7%) and four in the control group (8%) were excluded from the study because of one or more previous hospitalizations.
	R2	Low risk	There were missing datas in terms of non balanced participants between each questions. But the questions were analyised separately and the insufficient details is not enough to produce clinically relevant impact on the study .	High Risk	We excluded children who had incomplete data, such as missing answers to any of the multiple choice question in the survey. Children who completed the multiple choice component but did not answer the open-ended questions were still included.	Low Risk	A total of 71 participants had parental consent and assented to participate; no data were excluded from the final analysis.	Low Risk	No incomplete datas were detected.	High Risk	For reasons of nonappropriate age and outlier data, 8 children were excluded from final analysis (N = 131). Attritioan bias	Unclear risk	The exclusion of data was not mentioned.	Unclear risk	No incomplete datas were detected	Low Risk	All data were included in final analysis	High Risk	All the questionnaires are numbered, unfortunately questionnaires numbered 86-124 were not found. Thus we have only 85 filled questionnaires. Out of the 85 questionnaires, 28 of the questionnaires were not filled in fully, some only completed part 1 and not part 2 while others were scribbled on. Therefore, only 57 fully completed questionnaires were considered for this study.	Low Risk	Three children in the intervention group and four in the control group were excluded from the study because of one or more previous hospitalization.
Incomplete outcome data addressed (attrition bias) (long-term >6 weeks)	R1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	R2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Selective reporting (reporting bias)	R1	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported
	R2	Low Risk	All the outcomes were reported in a prespecifed way	Low Risk	All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported after a statistical analysis through Fisher Exact Test.	Low Risk	All outcomes were reported in a prespecified way.	Low Risk	All outocmes were reported in a prespecified way	Low Risk	All outcomes were reported in a prespecified way.	Low Risk	All outcomes were reported in prespecified way.	Low Risk	All outcomes were reported in prespecified way.	Low Risk	All outcomes were reported in prespecified way.
Other bias	R1	Low Risk	None	High Risk	Marion M. Aw is the clinical mentor for the medical students in charge of the Teddy Bear Hospital, Singapore. Ong Lynn and Chua Khoon Han were both involved in TBH as part of the organising committee during the course of this study.	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None
	R2	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None
Decided after discussion																					